H. T. KLECKNER.

METALLIC RAILWAY TIE AND FASTENING.

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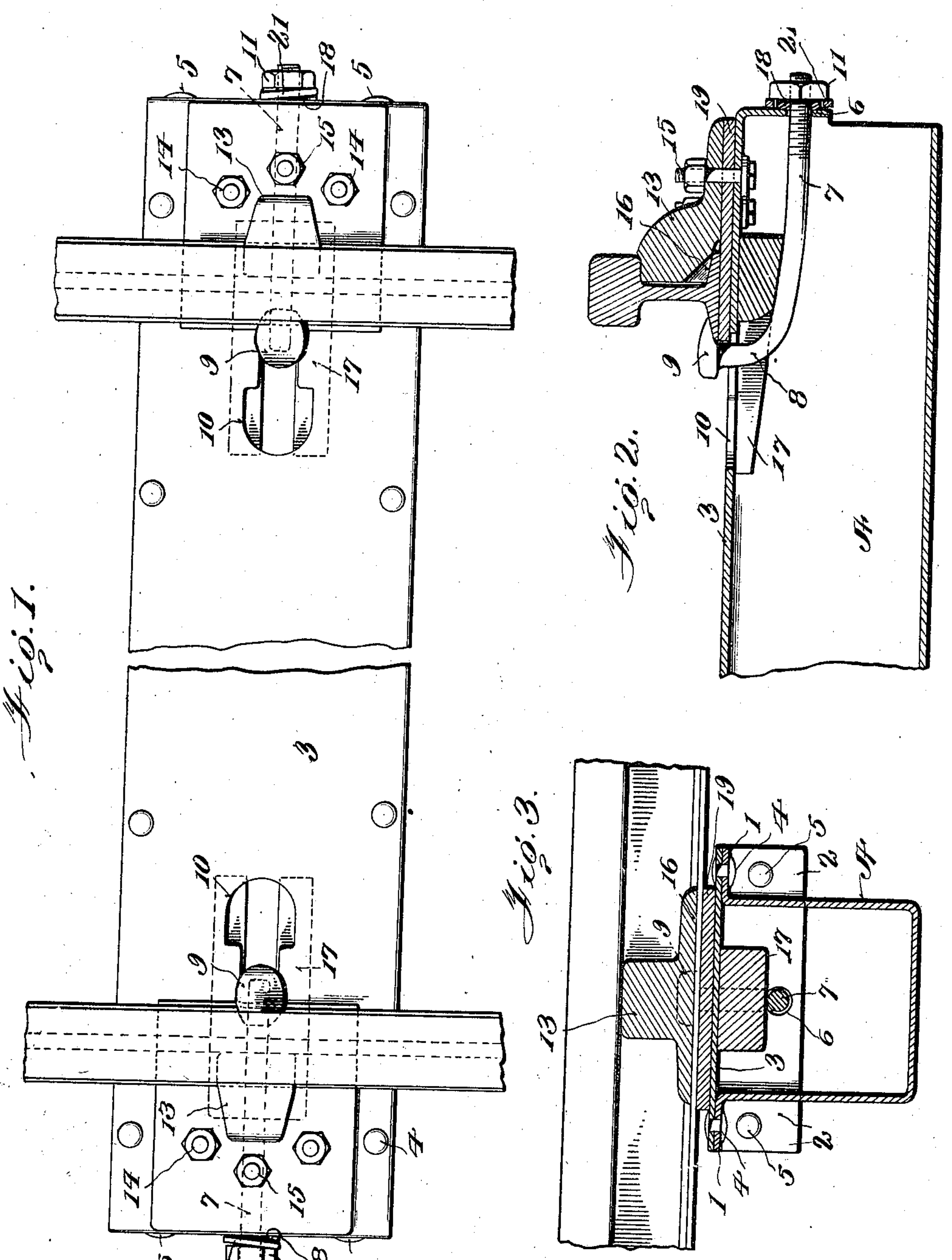
1,014,489. Patented Jan. 9, 1912.



Witnesses

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UNITED STATES PATENT OFFICE.

HARRY TILGHMAN KLECKNER, OF SHAMOKIN, PENNSYLVANIA, ASSIGNOR OF ONE-HALF TO IRVIN REBER, OF WINFIELD, PENNSYLVANIA.

METALLIC RAILWAY-TIE AND FASTENING.

1,014,489. Specification of Letters Patent. **Patented Jan. 9, 1912.**

Application filed October 21, 1911. Serial No. 655,869.

To all whom it may concern:

Be it known that I, HARRY T. KLECKNER, a citizen of the United States, residing at Shamokin, in county of Northumberland and State of Pennsylvania, have invented certain new and useful Improvements in Metallic Railway-Ties and Fastenings, of which the following is a specification.

My invention relates to an improvement in metallic railway ties and fastenings, and the object is to provide means for fastening the rail upon the tie, which tie is constructed preferably from two pieces of metal having the upper surface thereof provided with slots in which the fastening means is received for engaging the rail for fastening the rail upon the tie.

The invention consists in certain novel features of construction and combinations of parts which will be hereinafter described and pointed out in the claims.

In the accompanying drawings, Figure 1 is a top plan view; Fig. 2 is a longitudinal vertical sectional view of a portion of the tie; and Fig. 3 is a cross sectional view.

A represents the body of the tie, which is preferably U-shaped, and having outwardly extending flanges 1, 1, at the top thereof and outwardly projecting end flanges 2, 2. A top plate 3 is mounted upon the body A and is fastened to the flanges 1 by means of rivets or bolts 4, the extreme ends of the top plate being bent downwardly over the ends of the body A and fastened to the end flanges 2 by means of bolts or rivets 5. Openings 6 are formed in the ends of the plate, through which rods 7 extend. The rods are provided with angular extensions 8 which have heads 9 thereon. The angular extensions extend through slots 10 formed in the plate, and the heads are adapted to be drawn onto the rail flange. The other terminals of the rods are screw-threaded, and nuts 11 are mounted thereon for fastening and drawing the rods into engagement with the rail flange for fastening the rail to the tie. Clamping blocks 13 are mounted upon the tie plate 3, and are fastened thereto by means of bolts 14 and nuts 15. A recess 16 is formed in the clamping block for the reception of the flange of the rail, whereby the rail will be held in engagement with the tie.

Received between the rods 7 and the tie plate 3 are tapering slotted wedge blocks 17, the angular extension 8 of the bolts being received through the forked portion. The wedge block is intended to take up any loose play which might occur between the tie and rail. If the wedge block were omitted there would be considerable play allowed between the rail and flange, and the fastening rod, so that any vibration imparted to the rail would cause the bolt to move with the rail, and the rail is therefore moved up and down or vibrated upon the top of the tie, and it is to overcome such an objection that the wedge block is used. Furthermore, the wedge block, which is preferably of wood, acts as a cushion to allow for any strain imparted to the rail which might occur, and which occurs more especially upon the curves of a track.

From the foregoing it will be seen that when a rail is placed upon the tie and fitted beneath the clamping block 13, and the head of the rod 7 is drawn upon the opposite flange of the rail, by screwing the nut upon the bolt against the downwardly projecting end of the plate 3, the rail will be securely fastened to the tie, as the wedge or cushion block will fill the space between the bolt and the under surface of the plate 3 or top of the tie, so that the rail will be securely held upon the tie and prevented from moving or vibrating upon the tie.

It is my intention that insulated washers 18 shall be received between the nuts and downwardly projecting ends of the tie plate, so that in case of any electrical connection, the rail will be properly insulated. Of course, in such instances the clamping blocks 13 will be provided with insulated or wooden strips 19, which strips are located between the clamping blocks and top of the tie. Such blocks are also used when fish plates are to be applied.

It will be understood, of course, that the wedge blocks will be adjusted to suit the requirements when it is necessary to use the strips 19.

Suitable locking washers 21 are provided for fastening the nuts on the bolts.

Having fully described my invention, what I claim as new and desire to secure by Letters Patent, is:—

1. The combination with a hollow tie having slots therein, clamp blocks mounted on the tie for engaging the rail, of rods supported by the tie having angular extensions extend-

ing through the slots in the tie, means engaging the rods for drawing the angular extensions onto the rail flange for fastening the rail upon the tie, and wedging means received between the bolt and tie for preventing the vibration of the rail upon the tie.

2. The combination with a hollow tie having slots therein, clamp blocks mounted on the tie for engaging the rail, of rods supported by the tie having angular extensions extending through the slots in the tie, means engaging the rods for drawing the angular extensions onto the rail flange for fastening the rail upon the tie, tapering blocks received between the rods and tie for affording a cushion and preventing the vibration and movement of the rail upon the tie.

In testimony whereof I affix my signature, in the presence of two witnesses.

HARRY TILGHMAN KLECKNER.

Witnesses:
HARVEY MORGAN,
JOSEPH KLINGER.